

Enhancing Student Engagement through Interactive Learning Tools

AYUSH DEVKAR & VIDHI MANANI

Department of Information Technology,

SHAH AND ANCHOR KUTTCHI ENGINEERING COLLEGE MUMBAI,

INDIA

ayush.devkar241@sakec.ac.in

vidhi.manani24@sakec.ac.in

1. PROBLEM STATEMENT:

Despite the widespread availability of digital learning resources, many students continue to face challenges in grasping complex concepts, sustaining attention, and managing their academic responsibilities effectively. Traditional teaching methods often follow a one-size-fits-all approach, failing to accommodate differences in learning pace and individual understanding.

As a result, students frequently depend on rote memorization rather than conceptual learning, leading to increased academic stress and inconsistent study patterns.

Additionally, most existing learning systems do not account for variations in learning preferences, cognitive abilities, or time constraints. This lack of adaptability reduces student engagement and confidence.

Therefore, there is a need for a unified and intelligent learning system that can simplify content delivery, support continuous self-evaluation, and enable personalized study planning tailored to individual needs.

2. Abstract:

Students often encounter difficulties related to poor time management, unstructured revision strategies, and scattered learning resources, all of which negatively impact academic performance.

To address these issues, this project introduces *Smart Learning*, a desktop-based academic support system developed using Python and the Tkinter framework. The system integrates multiple functionalities, including personalized scheduling, organized note management, multimedia-based learning, and access to previous examination papers within a single platform.

The scheduling module generates customized weekly study plans by considering subject importance, difficulty level, and available study time. It also incorporates revision sessions, progress tracking, and task highlighting, with data stored persistently using JSON.

The platform further includes subject-wise note organization, video-based learning support, and structured access to past papers using Excel datasets.

By combining planning, revision, and resource management, the system encourages disciplined study habits, enhances conceptual understanding, and improves overall academic productivity.

3. INTRODUCTION:

Traditional education systems have long relied on textbooks, lectures, and examinations as the primary methods of knowledge delivery. While these approaches have been effective in the past, they often promote passive learning, where students receive information without actively engaging with it.

In today's digital environment, such methods are no longer sufficient to maintain student interest or ensure deep understanding. Despite increased access to online resources, many students still struggle with concept clarity, focus, and academic planning.

One of the major limitations of conventional systems is their inability to adapt to individual learning needs. Students with different learning speeds and abilities are treated uniformly, which often leads to disengagement and reduced academic confidence.

These challenges highlight the need for a more structured and personalized learning approach.

This project proposes *Smart Learning*, an integrated academic support system designed to improve study efficiency, conceptual clarity, and productivity by combining organization, accessibility, and adaptability within a single platform.

4. SDG Format

SDG 4: Quality Education

Why it fits BEST:

Your problem statement clearly focuses on:

- Difficulty in understanding complex concepts
- Lack of personalized learning
- Poor study habits & academic stress
- Need for adaptive, student-centered learning systems

SDG 4 aims to:

- Ensure inclusive and equitable quality education
- Promote effective learning outcomes
- Support personalized and lifelong

learning Your system directly contributes by:

- Personalized study planning (adaptive learning)
- Continuous self-assessment (better learning outcomes)
- Integrated learning (improves engagement & clarity)

SDG 9: Industry, Innovation and Infrastructure

Why it fits:

- Your system is a technology-driven solution (Python + GUI system)
- Introduces innovation in education systems
- Combines multiple tools into one integrated platform

This supports:

- Digital innovation
- Smart infrastructure for education

SDG 3: Good Health and Well-being (*Indirect but valid*)

Why it fits:

- Your problem mentions:
 - Academic stress
 - Lack of confidence
 - poor study habits

Your system helps by:

- Reducing stress (better planning)
- Increasing confidence (progress tracking)
- Improving mental well-being of students

Objective of Problem Statement

Main Objective

- To develop an integrated learning system that improves understanding, focus, and academic performance through personalized and adaptive learning support.

Specific Objectives

- ✓ To simplify complex learning content for better understanding
- ✓ To provide personalized learning experience based on student needs
- ✓ To enable continuous self-assessment and performance tracking
- ✓ To help students identify weak areas in real time
- ✓ To improve time management and study planning
- ✓ To reduce academic stress and dependency on rote learning
- ✓ To increase student engagement and learning consistency

5. LITERATURE REVIEW:

+	No.	Title	Author(s)	Publisher	Date Published	Key Summary
	1	LectureNote: Automatic Generation of Lecture Notes from Slide Based Video	Xu et al.	IEEE ICME	2019	Automated note generation using speech recognition and slide extraction; improves efficiency.
	2	NoteLink: A Point-and-Shoot-Linking interface between Student Handwriting and Lecture Video	Dadeo et al.	IEEE/ACM JCDL	2021	Connects handwritten notes to video timestamps. Enhances review efficiency and coordination.
	3	Researching and Supporting Student Note-taking: Building a Multimedia Note	IEEE TALE Researchers	IEEE Conference	2015	Multimedia and structured note-taking improves engagement and memory retention.
	4	Visual Summarization of Lecture Video Snippets for Enhanced Navigation	IEEE Researchers	IEEE Conference	2020	Segments lecture videos into structured sections for easier navigation and reduced cognitive load.
	5	Self-Regulated Learning in Technology-Enhanced Learning Environments	Computers & Education (Elsevier)	Comp. 2023	2015	Highlights importance of time management, planning, and self-monitoring in learning.
	6	Design and Implementation of a Time Management System	Zhao et al.	IEEE	2023	Personalized scheduling improves time management practices.

4. Existing Systems

Conventional education systems primarily depend on classroom instruction, textbooks, and examination-based assessments. Although digital learning tools have increased accessibility, most of them operate independently rather than as a cohesive system.

For instance, scheduling applications focus only on time planning, note-taking tools emphasize content organization, and video platforms provide lecture recordings without integration.

This fragmentation forces students to switch between multiple platforms, leading to inefficiency and disorganized learning experiences.

Furthermore, many of these tools lack personalization and fail to adjust based on individual study patterns or subject difficulty. They also do not provide structured revision mechanisms or effective progress tracking.

As a result, students often develop inconsistent study habits and struggle with time management despite having access to multiple digital tools.

5. Proposed System

To address the limitations of existing systems, this project introduces *Smart Learning*, a desktop-based academic support platform developed using Python and the Tkinter GUI framework.

The system integrates multiple learning functionalities into a single interface, including:

- Personalized study scheduling
- Subject-wise note organization
- Multimedia-based video learning
- Past paper access using structured datasets
- Progress tracking and task completion indicators

Unlike fragmented tools, this system provides a unified environment that combines planning, revision, and learning resources.

The scheduling feature generates customized study plans based on subject priority, difficulty, and available time, while automatically incorporating revision sessions.

Additionally, progress tracking and visual indicators help students monitor their performance, encouraging self-regulated learning behavior.

6. Methodology

The development of Smart Learning followed a modular and systematic approach influenced by research on structured educational technologies and time-management systems [1][5][6].

1. Requirement Analysis

The first phase involved identifying common student challenges such as poor time management, unstructured revision, and scattered learning resources.

These challenges align with findings in self-regulated learning research [5] and personalized scheduling systems [6]. Based on these observations, core system modules were defined: scheduling, notes, video integration, and past paper access.

2. System Design

The system was designed using a modular architecture where each feature functions independently but integrates seamlessly within the main dashboard.

The structured organization approach is influenced by multimedia note-taking research [3] and lecture structuring concepts [1][4]. The graphical interface was developed using Tkinter to ensure user-friendly navigation and accessibility.

3. Implementation

Scheduling Module:

- Accepts user inputs (subjects, study hours, difficulty level).
- Generates a weekly timetable algorithmically, supporting personalized planning [6].
- Automatically assigns revision slots based on structured scheduling principles [6].
- Highlights completed tasks to reinforce progress awareness [5].
- Displays a dynamic progress bar to support self-monitoring behavior [5].
- Stores data persistently using JSON to maintain study continuity.

Notes Management Module

- Allows subject-wise creation and storage of notes.
- Provides organized dashboard-based access.
- Ensures structured academic content management inspired by multimedia note research [3] and automated note structuring concepts [1].

Video Module

- Supports topic-based educational video access.
- Enhances conceptual clarity through multimedia learning, aligned with structured video segmentation research [4] and lecture-based content extraction approaches [1].

Past Paper Module

- Uses structured Excel datasets for organized storage.
- Enables filtering and retrieval of previous examination papers.
- Supports exam-oriented preparation through systematic academic resource organization [5].

4. Testing and Validation

The system was tested for:

- Accuracy of timetable generation [6]
- Data persistence functionality
- UI responsiveness
- Correct retrieval of past papers
- Progress tracking accuracy [5]

The testing phase ensured stability, usability, and reliability while maintaining alignment with structured educational system design principles.

7. Results and Analysis

The developed system demonstrated the following outcomes:

- Successfully generated structured weekly timetables based on user input, supporting personalized scheduling efficiency [6].
- Automated revision slot allocation improved study balance and reduced workload clustering [6].
- Task highlighting visually indicated completed activities, reinforcing self-monitoring and accountability [5].
- Progress bar effectively displayed completion percentage, supporting self-regulated learning behaviors [5].
- Notes module improved subject-wise organization, reflecting structured multimedia learning benefits [3].
- Video module enabled segmented and organized content access, consistent with structured lecture research [4].
- Past paper module allowed quick and structured retrieval of exam materials, improving exam readiness.
- Persistent JSON storage ensured that user data remained saved across sessions, enhancing usability and continuity.

Overall, the system reduced dependency on multiple platforms and improved structured academic planning by integrating features inspired by structured note generation [1], multimedia organization [3], video segmentation [4], self-regulated learning [5], and personalized scheduling systems [6].

- Main dashboard
- Generated timetable
- Highlighted completed task
- Progress bar
- Notes interface
- Past paper search interface

These screenshots visually validate the functional implementation and practical effectiveness of the system.

8. Conclusion and Future Scope

Conclusion

The Smart Learning system successfully integrates personalized scheduling, structured note management, multimedia learning support, and past paper access within a single desktop-based platform. By addressing common challenges such as time mismanagement, fragmented resources, and unstructured revision, the system promotes disciplined study habits and improved academic organization.

The modular architecture ensures scalability and usability, while features such as revision allocation, progress tracking, and persistent storage enhance learning efficiency. The project demonstrates how a technology-driven academic support system can improve productivity and structured learning practices.

Future Scope

The system can be further enhanced by:

- Developing a cloud-based or web version for remote accessibility
- Adding AI-based recommendation systems for personalized study plans
- Implementing real-time performance analytics
- Introducing collaborative note-sharing features
- Creating a mobile application version
- Adding adaptive learning mechanisms based on user performance data

These enhancements can make the system more intelligent, scalable, and widely accessible.

REFERENCES:

- [1] C. Xu, R. Wang, S. Lin, X. Luo, B. Zhao, L. Shao, and M. Hu, "Lecture2Note: Automatic Generation of Lecture Notes from Slide-Based Educational Videos," in *Proceedings of the IEEE International Conference on Multimedia and Expo (ICME)*, 2019. Available: <https://ieeexplore.ieee.org/document/8784916/>
- [2] S. Dodson, R. J. Srinivasa, K. Seo, D. Yoon, and S. Fels, "NoteLink: A Point-and-Shoot Linking Interface between Students' Handwritten Notebooks and Instructional Videos," in *Proceedings of the ACM/IEEE Joint Conference on Digital Libraries (JCDL)*, 2021. Available: <https://ieeexplore.ieee.org/document/9651906/>
- [3] M. C. O'Connor, J. B. Staudt, and K. A. Cunningham, "Researching and Supporting Student Note-Taking: Building a Multimedia Note-Taking App," in *IEEE International Conference on Teaching, Assessment, and Learning for Engineering (TALE)*, 2015. Available: <https://ieeexplore.ieee.org/document/7386015/>
- [4] Y. Liu, Z. Zhao, and H. Wang, "Visual Summarization of Lecture Video Segments for Enhanced Navigation," in *IEEE International Conference on Multimedia Systems*, 2020. Available: <https://ieeexplore.ieee.org/document/9327964/>
- [5] Z. Li, H. Chen, and Q. Zhang, "Summarizing Lecture Videos by Key Handwritten Content Regions," in *IEEE International Conference on Image Processing (ICIP)*, 2019. Available: <https://ieeexplore.ieee.org/document/8892942/>
- [6] W. Zhao, H. Harb, M. Muntaser, and P. Bernacki, "Design and Implementation of a Time Management Self-Help Mobile App for College Students," in *IEEE Integrated STEM Education Conference (ISEC)*, 2023. Available: <https://ieeexplore.ieee.org/document/10402177/>

- [7] D. Byrne, P. Wilkins, G. J. F. Jones, A. F. Smeaton, and N. E. O'Connor, "Measuring the Impact of Temporal Context on Video Retrieval," in *Proceedings of the 2008 International Conference on Content-Based Image and Video Retrieval*, 2008. Available: <https://ieeexplore.ieee.org/>
- [8] G. Awad, C. G. M. Snoek, A. F. Smeaton, and G. Quénot, "TRECVID Semantic Indexing of Video: A 6-Year Retrospective," *ITE Transactions on Media Technology and Applications*, vol. 4, no. 3, pp. 187–208, 2016. Available: <https://ieeexplore.ieee.org/>
- [9]
- [10] A. Delaye and C. L. Liu, "Text/Non-Text Classification in Online Handwritten Documents with Conditional Random Fields," in *Chinese Conference on Pattern Recognition*, 2012. Available: <https://ieeexplore.ieee.org/>
- [11] A. Delaye and C. L. Liu, "Contextual Text/Non-Text Stroke Classification in Online Handwritten Notes with Conditional Random Fields," *Pattern Recognition*, vol. 47, no. 3, pp. 959–968, 2014. Available: <https://ieeexplore.ieee.org/>
- [12] D. Castro, B. L. D. Bezerra, and M. Valença, "Boosting the Deep Multidimensional Long-Short-Term Memory Network for Handwritten Recognition Systems," in *16th International Conference on Frontiers in Handwriting Recognition (ICFHR)*, 2018. Available: <https://ieeexplore.ieee.org/>
- [13] S. Dodson, I. Roll, N. M. Harandi, S. Fels, and D. Yoon, "Weaving Together Media, Technologies, and People: Students' Information Practices in Flipped Classrooms," *Information and Learning Sciences*, vol. 120, no. 7/8, pp. 519–540, 2019. Available: <https://ieeexplore.ieee.org/>
- [14] B. Dorn, L. B. Schroeder, and A. Stankiewicz, "Piloting TrACE: Exploring Spatiotemporal Anchored Collaboration in Asynchronous Learning," 2015. Available: <https://ieeexplore.ieee.org/>
- [15] J. Adcock, M. Cooper, L. Denoue, H. Pirsiavash, and L. A. Rowe, "TalkMiner: A Lecture Webcast Search Engine," in *Proceedings of the 18th ACM International Conference on Multimedia*, 2010. Available: <https://ieeexplore.ieee.org/>

[8] G. Awad, C. G. M. Snoek, A. F. Smeaton, and G. Quénot, "TRECVID Semantic Indexing of Video: A 6-Year Retrospective," *ITE Transactions on Media Technology and Applications*, vol. 4, no. 3, pp. 187–208, 2016.

Available: <https://ieeexplore.ieee.org/>

[9]

[10] A. Delaye and C. L. Liu, "Text/Non-Text Classification in Online Handwritten Documents with Conditional Random Fields," in *Chinese Conference on Pattern Recognition*, 2012. Available:

<https://ieeexplore.ieee.org/>

[11] A. Delaye and C. L. Liu, "Contextual Text/Non-Text Stroke Classification in Online Handwritten Notes with Conditional Random Fields," *Pattern Recognition*, vol. 47, no. 3, pp. 959–968, 2014. Available:

<https://ieeexplore.ieee.org/>

[12] D. Castro, B. L. D. Bezerra, and M. Valença, "Boosting the Deep Multidimensional Long-Short-Term Memory Network for Handwritten Recognition Systems," in *16th International Conference on Frontiers in Handwriting Recognition (ICFHR)*, 2018. Available:

<https://ieeexplore.ieee.org/>

[13] S. Dodson, I. Roll, N. M. Harandi, S. Fels, and D. Yoon, "Weaving Together Media, Technologies, and People: Students' Information Practices in Flipped Classrooms," *Information and Learning Sciences*, vol. 120, no. 7/8, pp.

519–540, 2019. Available: <https://ieeexplore.ieee.org/>

[14] B. Dorn, L. B. Schroeder, and A. Stankiewicz, "Piloting TrACE: Exploring Spatiotemporal Anchored Collaboration in Asynchronous Learning," 2015.

Available: <https://ieeexplore.ieee.org/>

[15] J. Adcock, M. Cooper, L. Denoue, H. Pirsiavash, and L. A. Rowe, "TalkMiner: A Lecture Webcast Search Engine," in *Proceedings of the 18th ACM International Conference on Multimedia*, 2010. Available:

<https://ieeexplore.ieee.org/>

[16] P. Borlund, "The IIR Evaluation Model: A Framework for Evaluation of Interactive Information Retrieval Systems," *Information Research*, vol. 8, no. 3, 2003. Available: <https://ieeexplore.ieee.org/>

[17] L. Barker, C. L. Hovey, J. Subhlok, and T. Tuna, "Student Perceptions of Indexed, Searchable Videos of Faculty Lectures," in 44th Annual Frontiers in Education Conference (FIE), 2014. Available: <https://ieeexplore.ieee.org/>

[18] S. D. Sorden and J. L. Ramírez-Romero, "Collaborative Learning, Social Presence and Student Satisfaction in a Blended Learning Environment," in IEEE International Conference on Advanced Learning Technologies, 2012.
Available: <https://ieeexplore.ieee.org/>

[19] T. Tuna et al., "Topic-Based Segmentation of Classroom Videos," in 45th Annual Frontiers in Education Conference (FIE), 2015. Available: <https://ieeexplore.ieee.org/>

[20] T. Tuna et al., "Indexed Captioned Searchable Videos: A Learning Companion for STEM Coursework," *Journal of Science Education and Technology*, vol. 26, no. 1, pp. 82–99, 2017. Available: <https://ieeexplore.ieee.org/>

[21] R. S. Koka, F. N. Chowdhury, M. R. Rahman, T. Solorio, and J. Subhlok, "Automatic Identification of Keywords in Lecture Video Segments," in IEEE International Symposium on Multimedia, 2020.
Available: <https://ieeexplore.ieee.org/>

